

MOULAY ALI SAKURAI EL IDRISSE

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Summary

Systems-focused software engineer building at the intersection of machine learning and security. Production experience designing LLM pipelines, optimising database performance at scale, and implementing post-quantum cryptographic protocols. First Class Honours in Cyber Security, 1st place in 42 Tokyo's Piscine, and a journal paper under peer review.

EDUCATION

Nottingham Trent University

Bachelor's, CyberSecurity

Sep 2022 - May 2025

Nottingham, UK

- **GPA:** 1st Class Honours

- **Achievements:** Department of Computer Science Project Prize in BSc (Hons) Cyber Security

42 Tokyo

Software engineering

Sep 2025 - Present

Tokyo, Japan

- **Achievements:** 1st place in the entrance exam(Piscine)

PROFESSIONAL EXPERIENCE

DIGITAL GRID Corporation

Backend Engineer Intern

Feb 2026 - Present

Tokyo, Japan

- Designed and built an enterprise conversational AI system using LangGraph, implementing advanced retrieval strategies, hybrid vector search, hierarchical document clustering, and a knowledge graph with entity-relationship traversal, which surpassed the previous system's answer accuracy by ~20% through automated evaluation benchmarking.
- Strengthened backend security and performance by enforcing JWT-based authentication, role-based access control, and input sanitisation across all API endpoints, while reducing database query overhead by ~95% through query optimisation on tenant-isolated services.
- Configured the full DevOps pipeline, containerized deployment with Docker Compose, asynchronous task processing, database migration tooling, and a real-time monitoring stack with Prometheus and Grafana, enabling faster releases and continuous system observability
- Supported delivery of 81 requirements across a 12-phase Agile roadmap by coordinating tasks, tracking progress, and collaborating with a cross-functional team to ensure timely completion

Tokyo ICT(JICA)

Full Stack Engineering Intern

Jul 2023 - Oct 2023

Tokyo, Japan

- Developed a multilingual speech conversion system using Azure Cognitive Services, leveraging Speech-to-Text and Text-to-Speech APIs to provide real-time language processing, which enhanced communication efficiency for users
- Built scalable backend services using Django, Node.js, and PostgreSQL, reducing runtime by 30% and optimizing performance for multilingual voice applications.
- Received recognition from the Representative Director after delivering a solution that reduced translation errors and streamlined the workflow, demonstrating effective collaboration with the team

Brando SARL

Junior Developer Intern

Nov 2022 - Feb 2023

Marrakech, Morocco

- Developed responsive web applications using JavaScript and RESTful APIs to automate workflows and reduce manual overhead.
- Collaborated with a team using Agile methodologies and Git to deliver client-focused solutions within tight deadlines, enhancing client satisfaction and project delivery speed

PUBLICATIONS & MANUSCRIPTS

- Sakurai El Idrissi, M. A., Otuka, R., Nwajana, A. "Integrating Machine Learning Fraud Detection and Hybrid Post-Quantum Cryptography in a Secure Trading Platform." Technologies (MDPI), Special Issue "Disruptive Technologies: Big Data, AI, IoT, Games, and Mixed Reality". Manuscript under review: Combines ML ensemble methods for financial fraud detection with hybrid post-quantum cryptographic protocols for secure trading infrastructure.

PROJECTS & OUTSIDE EXPERIENCE

ML-Driven Trading Fraud Detection with Post-Quantum Cryptography

Nottingham Trent University

Aug 2024 - May 2025

- Built fraud detection models using scikit-learn and Python, achieving 89% accuracy on simulated trading data through ensemble methods and feature engineering on time-series transaction patterns.
- Integrated generative AI for anomaly detection, identifying outlier trading behaviours that rule-based systems missed.
- Evaluated quantum-resistant encryption (CRYSTALS-Dilithium, SPHINCS+) for securing trading platform communications.
- Published findings in MDPI Technologies journal (under review).

Options Pricing Engine in C++

Dec 2025 - Mar 2026

- Implemented Black-Scholes analytical pricing and Monte Carlo simulation for European call/put options, validating numerical convergence against closed-form solutions.
- Built modular OOP architecture with separate classes for payoff functions, pricing models, and random number generation, supporting extensibility to exotic options.
- Optimised Monte Carlo path generation achieving ~10 million simulations/second through cache-friendly memory layout and variance reduction techniques (antithetic variates), reducing required sample size by 3x for equivalent accuracy.
- Computed option Greeks via finite difference approximation.

Deadlock-Free Concurrent Scheduling System in C

Oct 2024 - Apr 2025

42 Tokyo

- Solved Dijkstra's dining philosophers problem with zero deadlocks and zero data races, managing up to 200 concurrent threads sharing 200 mutex-protected resources.
- Implemented precise timing control with microsecond-granularity scheduling to prevent philosopher starvation under high-contention scenarios.
- Built thread-safe logging system for real-time state monitoring without mutex contention on output.

TECHNICAL SKILLS

- **Languages:** C, C++, Python, SQL, Go
- **Framework & Tools:** Django, LangChain, LangGraph, Docker, Git, FastAPI, Scikit-learn, Grafana
- **Cloud & Data:** AWS, PostgreSQL, ChromaDB, Rest APIs
- **AI/ML & Others:** Retrieval-Augmented Generation, Generative AI, Vector Databases, Prometheus, Loki, Agile, API Security